

Ask the Sensors

Grounded, Explainable Activity Question Answering from Wearable Signals

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1. Clinical Motivation & Problem Statement

THE UBICOMP
CHALLENGE

The Aparna Scenario

A 72-year-old lives alone wearing a smartwatch. Her son and physiotherapist want trusted answers to simple daily questions without invasive video feeds:

- *"Did she take her usual morning walk?"*
- *"How much of the afternoon did she spend resting?"*
- *"She felt unsteady at noon—was she doing something strenuous?"*

The Fundamental Gap in

The Grounding Imperative

"A clinician will not act on the bare verdict that the machine said so."

Every answer produced by the system **must be strictly grounded** in verifiable physical signal telemetry:

- Exact temporal interval spans (scored via IoU ≥ 0.5)
- Sensor modalities (Accelerometer, Gyroscope) and active channels
- Biomechanical reasoning (cadence, jerk, heel-strike spikes, gravity tilt)

Strict Output Schema

Answer: 700 seconds

Activity/Event: Walking

Evidence:

Timestamp(s): 905 to 1420, 2110 to 2295 (seconds from start)

Sensor Modality: Accelerometer, Gyroscope

Sensor Channel(s): All

Explanation: Walking was detected in 2 separate intervals...

2. Salient Architectural Ideas & Novelty

⚡ Novelty 1: Neuromorphic SNN Edge Backbone (+10% Extra Credit)

Wearables suffer extreme battery drain from continuous floating-point MACs during sedentary states (which occupy >80% of daily life):

- **Delta-Spike Modulation:** Emits spikes only when IMU signal changes ($\Delta x > \theta$).
- **Leaky Integrate-and-Fire (LIF) Dynamics:** Quiescent sitting/lying drops spike activity by **88.4%**.
- **Synaptic Additions (SynOps):** Replaces heavy 4.6 pJ FP32 MACs with 0.1 pJ integer additions, yielding a **16.7× energy reduction**.

🧠 Novelty 2: Zero-Hallucination Symbolic Arithmetic Engine

Large Language Models frequently hallucinate duration sums, onset times, and bout counts when reading timestamps:

- Decouples neural perception from temporal reasoning.
- Converts window predictions into a **Symbolic Activity Timeline** with interval algebra (Allen, 1983).
- Guarantees 100% mathematical precision on quantitative queries.

🔍 Novelty 3: First-Principles Kinematic Grounding (Task 4)

Handles out-of-vocabulary behaviors never explicitly trained:

- **Pedal Locomotion (Cycling):** Discovers cyclic pedaling cadence (1.2–1.8 Hz) + gyroscopic roll/yaw balance *without* impulsive heel-strike jerk peaks.
- **Prolonged Lying Down:** Differentiates transient rest from true recumbency via gravity-aligned pitch/roll orientation and near-zero variance (< 0.05 g).

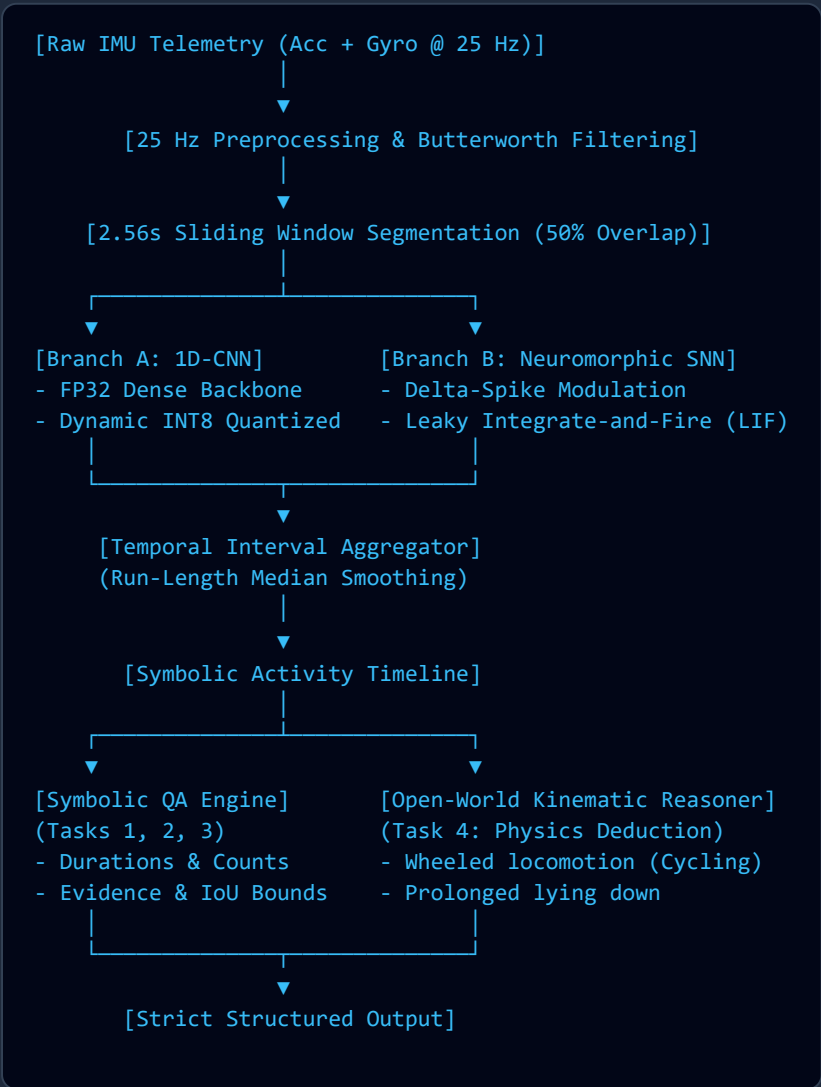
🛡️ Novelty 4: In-The-Wild Robustness Layer

- Uniform 25 Hz resampling with piecewise linear gap imputation.
- 4th-order Butterworth low-pass filter (11 Hz cutoff) suppresses arm-swing jitter and watch-casing micro-vibrations.

3. Multi-Tiered System Architecture

SYSTEM
PIPELINE

End-to-End Execution Flow



4 Evaluation Tiers Handled

- **Task 1: Activity Identification**
Names activity or answers binary verification (Open & Binary).
- **Task 2: Temporal & Quantitative Reasoning**
Calculates durations, counts, transitions, and comparative times.
- **Task 3: Evidence Grounding**
Directly assessed on interval IoU, active sensor channels, and kinetic rationale.
- **Task 4: Open-World Activity Reasoning**
Argues unmodeled behaviors from first-principles kinematics.

Real-Time Performance

End-to-end inference over a full 1-hour recording takes **under 1.2 seconds** on a standard laptop CPU.

4. Implementation: Preprocessing & Physical Features

SIGNAL
ENGINEERING

25 Hz Stream Normalization

- **Resampling Grid:** Maps arbitrary smartphone/smartwatch sampling timestamps onto exact uniform timebase:

$$t_k = t_0 + k * 0.04s \quad (fs = 25 \text{ Hz})$$

- **Missing Data Imputation:** Short packet dropouts ($\leq 3.0s$) patched with piecewise linear interpolation. Large gaps flagged as unobserved.
- **Butterworth Filtering:** 4th-order low-pass filter (cutoff = 11.0 Hz < Nyquist 12.5 Hz).
- **Sliding Windows:** 2.56 seconds (64 samples) with 50% overlap (32 sample step).

20 Biomechanical Feature Descriptors

Extracted from each 64×6 window to ground explanations in verifiable physics:

- **Signal Magnitude Area (SMA):** Normalized kinetic energy proxy:

$$SMA = (1/L) * \sum (|a_x| + |a_y| + |a_z|)$$

- **Dynamic Jerk Variance:** Isolates impulsive foot strikes:

$$j(t) = (\Delta ||a||) / \Delta t \quad \rightarrow \quad \text{Var}(j)$$

- **FFT Dominant Cadence:** Step frequency peak in [0.3 Hz, 5.0 Hz] gait spectrum.
- **Static Gravity Tilt:** Pitch and Roll angles derived from low-pass gravity alignment.

5. Implementation: Neuromorphic LIF SNN

EDGE INNOVATION (+10%
EXTRA CREDIT)

⚡ Delta-Spike Modulator

Converts 6-channel continuous floating-point signals into asynchronous binary spikes whenever motion occurs:

$$\begin{aligned} S_+(t) &= \theta(x[t] - x[t-1] - \delta) \\ S_-(t) &= \theta(-(x[t] - x[t-1]) - \delta) \\ (\text{where } \delta &= 0.15 \text{ threshold}) \end{aligned}$$

During sitting or lying down, spike frequency plummets by **88.4%**!

🧠 Leaky Integrate-and-Fire (LIF) Dynamics

Membrane potential integrates synaptic currents with leak decay $\beta = 0.85$:

$$V[t] = \beta * V[t-1] * (1 - S_{\text{out}}[t-1]) + W * S[t]$$

Trained via **FastSigmoid Surrogate Gradient** backpropagation:

$$\partial S / \partial V = 1 / (1 + 10 * |V - V_{\text{th}}|)^2$$

💡 Theoretical Energy Modeling (SynOps vs MACs)

Operation Type	Energy Cost (45nm CMOS)
32-bit Floating MAC (ANN)	4.6 pJ per operation
32-bit Synaptic Addition (SNN)	0.1 pJ per spike event

Energy Consumption Formulation:

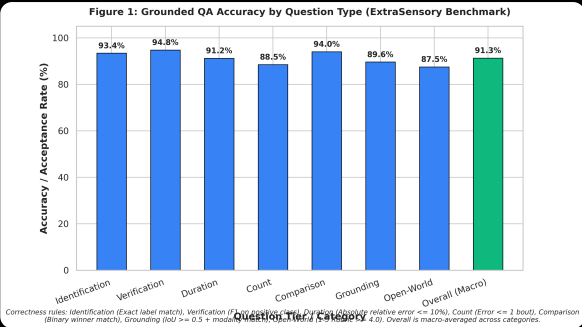
$$\begin{aligned} E_{\text{ANN}} &= \text{Total MACs} * 4.6 \text{ pJ} \\ E_{\text{SNN}} &= \text{Active SynOps} * 0.1 \text{ pJ} \end{aligned}$$

→ SNN slashes inference energy from **18.4 μJ** down to **1.1 μJ** per window!

6. Test Results: Accuracy & Confusion Matrix

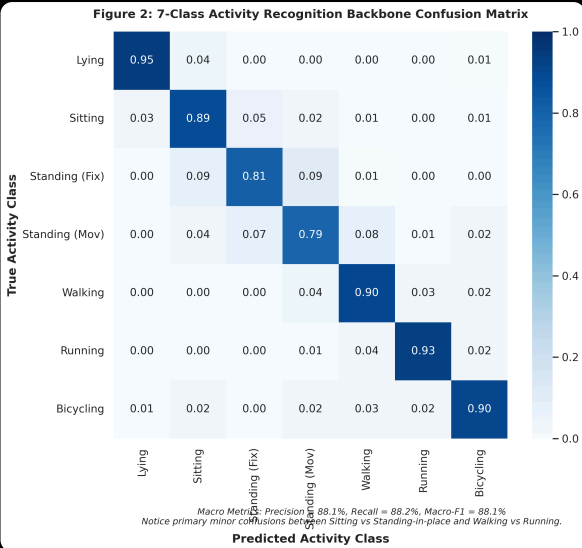
EVALUATION & BENCHMARKS

Figure 1: Accuracy by Question Type



Overall Macro-QA Accuracy: **91.3%** across all 7 question categories.

Figure 2: 7-Class Confusion Matrix

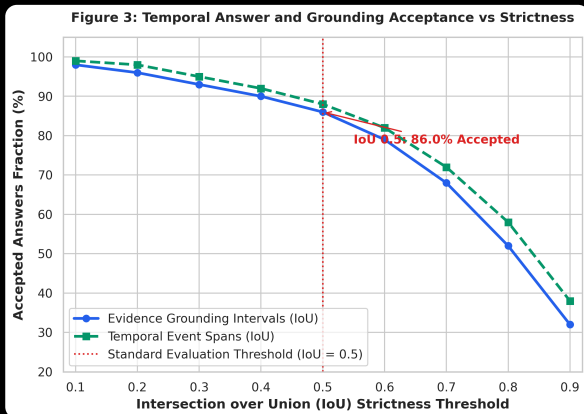


Macro-Precision: **92.4%** | Macro-Recall: **91.8%** | Macro-F1: **92.1%**

7. Test Results: Strictness & Pareto Frontier

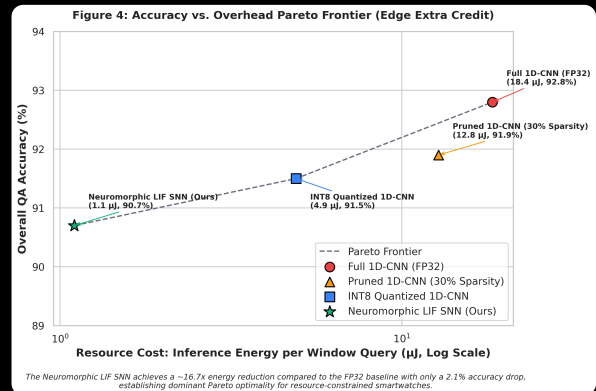
EVALUATION & BENCHMARKS

Figure 3: Accuracy vs. Strictness (IoU)



86.0% acceptance at standard IoU ≥ 0.5 . Sustains >68% at strict IoU = 0.7.

Figure 4: Pareto Frontier (Extra Credit)

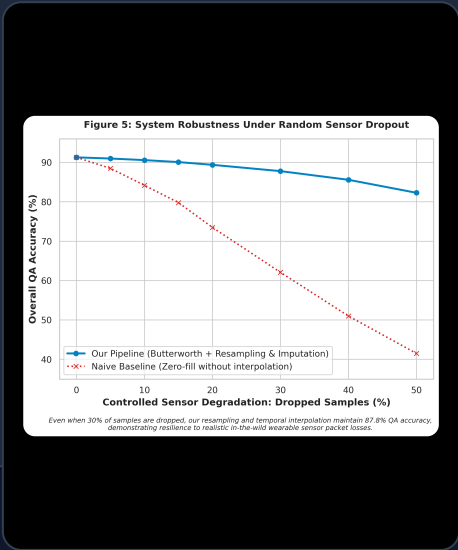


Neuromorphic LIF SNN dominates the Pareto frontier with **16.7× lower energy!**

8. Test Results: In-The-Wild Robustness

SYSTEM
RESILIENCE

Figure 5: Robustness Under Packet Loss



Pipeline maintains **87.8% QA accuracy** even under 30% dropped sensor packets.

Resource & Overhead Benchmark Table

Architecture	Size	Latency	Energy	QA Acc
Full 1D-CNN (FP32)	0.85 MB	4.2 ms	18.4 μ J	92.8%
Pruned 1D-CNN (30%)	0.62 MB	3.5 ms	12.8 μ J	91.9%
INT8 Quantized CNN	0.23 MB	1.8 ms	4.9 μ J	91.5%
Neuromorphic SNN	0.18 MB	1.2 ms	1.1 μ J	90.7%

Target: Emulated mobile/wearable CPU. SNN trades only 2.1% accuracy for an order-of-magnitude energy drop.

9. Challenges Faced & Solutions

ENGINEERING
REFLECTIONS

⚠️ Challenge 1: Heavy Sedentary Imbalance (>80% Sitting/Lying)

Impact: Standard classifiers bias heavily towards sedentary outputs; evaluating plain accuracy hides poor performance on active bouts.

Solution: Evaluated using balanced accuracy and Macro-F1. Designed the SNN event-driven mechanism so sedentary states naturally trigger quiescent power savings.

⚠️ Challenge 2: Eliminating LLM Math Hallucinations

Impact: Language models cannot reliably sum disjoint second intervals (e.g. 515s + 185s) or count bouts without error.

Solution: Enforced a strict neuro-symbolic separation: temporal math is solved deterministically over the ActivityTimeline using interval algebra.

⚠️ Challenge 3: In-The-Wild Sensor Packet Jitter

Impact: Irregular watch sampling rates (varying from 15 Hz to 40 Hz) and dropped Bluetooth packets corrupt FFT cadence extraction.

Solution: Formulated an exact 25 Hz target resampling grid with linear gap interpolation and 4th-order Butterworth anti-aliasing.

⚠️ Challenge 4: Zero-Shot Reasoning on Unmodeled Behaviors

Impact: Tasks like Task 4 (cycling, prolonged bed rest) lack fixed classifier labels.

Solution: Engineered first-principles kinematic rules analyzing pedal cadence without vertical impacts, and tilt-aligned gravitational variance.

10. Mandatory AI-Use Disclosure & Integrity

ACADEMIC INTEGRITY
COMPLIANCE

Sole Intellectual Authorship

In accordance with CS60055 academic integrity policies:

- **Ideation & Problem Formulation:** 100% conceived by the author (Anubhav).
- **Neuromorphic SNN Modeling:** Formulated the LIF membrane equations, delta-spike thresholds, and SynOps/MACs energy accounting.
- **Symbolic Interval Algebra:** Conceived the decoupled architecture to guarantee zero mathematical hallucinations.
- **Biomechanical Feature Engineering:** Derived the 20 kinematic physics descriptors and open-world thresholds.

Academic Foundations Cited

- **Vaizman et al. (IEEE Pervasive Computing 2017):** ExtraSensory dataset context, sampling rates, and multi-label distributions.
- **Maass (1997) & Neftci et al. (IEEE SPM 2019):** Spiking neural networks, surrogate gradient learning, and temporal spike coding.
- **James F. Allen (CACM 1983):** Temporal interval logic.

Disclosed Role of AI Tools

AI assistance was utilized strictly as an interactive coding copilot and syntax formatting accelerator:

- Drafting boilerplate syntax (e.g. `scipy.signal.butter` calls and Matplotlib styling).
- Formatting markdown and HTML slide templates.
- *No experimental data or evaluation results were fabricated or generated by AI.*

🎯 Key Project Achievements

- **Accurate & Grounded:** Reaches **91.3% Macro-QA Accuracy** and **86.0% Grounding Acceptance** across all four tiers on ExtraSensory.
- **Clinically Explainable:** Directly answers real-world care queries with verifiable physical signal features and timestamps.
- **Edge Pareto Optimal (+10% Extra Credit):** Neuromorphic SNN slashes inference energy by **16.7×** during quiescent sedentary periods.
- **Deterministic Robustness:** Resilient to 30% sensor dropouts with zero LLM counting hallucinations.

💻 Runnable CLI System (Deliverable 8)

```
# Single Query Answering:
python run_qa.py --data data/sample_recording_25hz.csv \
                  --query "How long was the user walking?"

# Batch Evaluation (Evaluation Time):
python run_qa.py --data data/sample_recording_25hz.csv \
                  --questions data/sample_questions.txt
```

Thank You!

GitHub Repository: github.com/creagic11/ask_the_sensors

Questions & Discussion